

Data Types

When it comes to Data Types, the list is huge, as there are various ways of classifying data, according to different criteria.

Some types we will glance over are:

Quantitative Data - Information that can be measured, counted, and expressed in numerical form. Usually represents numbers and measurable values.

Qualitative Data - Descriptive, and non-numeric information, which usually describes characteristics, such as gender, color, etc.

Structured Data - Information which has a fixed skeleton, or you can say, follows a particular structure/predefined format. Very useful for traditional relational databases.

Unstructured Data - Raw Information which is not constrained by any structure or format. Makes life hard for traditional SQL users and analysts. Some examples can be images, videos, pdfs, etc.

Semi-Structured Data - A mix of both structured, and unstructured data. A classic example of this is a JSON file - all JSON files have similar format, but the content inside the JSON file can differ from one file to another.

Some other types:

Numerical Data - Data represented using numbers.

Discrete Data - Usually concrete separate values, for example 1,2,3 are discrete numbers.

Continuous Data - Values that can take any value within a particular range, for example, 1.0,1.01,1.02.

Nominal Data - Categorical data with no inherent order whatsoever.

Ordinal Data - Categorical data with a meaningful order.

Binary Data - Data with only two possible values.

Text Data - Data represented as natural language.